

Christelle Seri

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EXPERIENCE

ERCOT | Application Developer II

Austin, Texas

July 2023 - Present

- Primary developer for a **\$2.5M** power grid communications project owning end-to-end backend architecture and database design
- Led backend performance optimization efforts by tuning database queries to improve response times and implementing **pagination** and **indexing**, reducing CPU saturation during load testing
- Designed and implemented **20+ SQL database tables** with constraints, procedures and indexing strategies
- Added automated **unit and integration tests** covering messaging success and error scenarios, improving reliability and reducing regression risk
- Developed **OpenAPI** specifications for core messaging APIs, improving documentation quality and API maintainability
- Contributed to GridGeo, a **React** and **Java**-based single page application used to visualize the Texas power grid in real time

Power Grid Communications Platform Rewrite (POC)

- Rebuilt the core functionality of a legacy Servlet/JDBC application as a **Spring Boot** and **Hibernate** proof of concept to evaluate scalability and maintainability
- Implemented REST APIs with validation, pagination, **LDAP authentication**, CSV export and global exception handling to demonstrate feature parity with the legacy application

PROJECTS

Event Driven Notification System

[Github](#)

- Built a distributed notification system using two **Spring Boot microservices** communicating asynchronously through **RabbitMQ** with a transactional outbox pattern for reliable event publication
- Improved messaging reliability through **retry with exponential backoff**, **dead-letter queues**, and end-to-end integration testing using **Mockito**, **MockMvc**, and **Testcontainers**
- Containerized the microservices with **Docker Compose** and deployed the application to an **AWS EC2** instance

EDUCATION

University of Texas at Austin

August 2024 - May 2026

Masters of Data Science

GPA: 3.73

University of Illinois at Urbana-Champaign

August 2019 - May 2023

Bachelor of Science in Electrical Engineering with Honors

GPA: 3.78

SKILLS

- **Programming Languages:** Java, SQL, Python, JavaScript
- **Databases:** Oracle, PostgreSQL
- **Frameworks:** Spring Boot, Spring Security, Hibernate, JUnit, Mockito, Testcontainers, React
- **Tools & Platforms:** Docker, RabbitMQ, Flyway, Git, Apache Tomcat, Maven